ARTIN–REES AND THE ACL CERTIFICATE: AN ELEMENTARY ACCOUNT

Status. An elementary version of `artin-rees-acl.tex`: everything is stated with polynomials and Laurent polynomials, without sheaves or cohomology. All steps are checked by Unit (`unit/sag/artin-rees`, `unit/sag/acl-zmt`); file names are given in parentheses. "Merely" means: under propositional truncation (the object exists, but is not part of the output).

1. The two statements

A ring R is *CFN* if (D) membership $v \in \operatorname{span}(u_1, \dots, u_k)$ is decidable, (C) the linear relations $\sum \lambda_i u_i = 0$ between finitely many elements are generated by finitely many of them, and (N) R is noetherian in the sense of Coquand–Persson. Discrete fields, $\mathbb{Z}$, polynomial rings, quotients and localisations of CFN rings are CFN; so is $A = k[b][a]$ for a discrete field k.

For $I = (a, b)$ and a submodule P, define $I^0 P = P$ and: $x \in I^{m+1}P$ if $x = ax_1 + bx_2$ with $x_1, x_2 \in I^m P$.

Theorem 1 (Artin–Rees)**.** *Let R be CFN, $I = (a, b)$, $N = \operatorname{span}(u_1, \dots, u_p) \subseteq R^r$. There merely is c such that, for every m: if $x \in I^{m+c}R^r$ and $x \in N$, then $x = \sum_j \lambda_j u_j$ with every $\lambda_j \in I^m$.*

In words: an element of N that is very divisible by I is a combination of the generators of N with coefficients that are almost as divisible, the loss c being independent of m (`ArtinRees`, `PowSpan`).

Theorem 2 (ACL)**.** *Let A be a CFN gcd domain, $a, b \in A$, and*

$$p = x + af + bg, \qquad q = y + au + bv \qquad (f, g, u, v \in A[x, y]).$$

In every A-algebra L containing x, y with $p(x, y) = q(x, y) = 0$ there merely are $r, t \in L$ such that, for $s = 1 + ar + bt$, each of s, sx, sy is a root of a monic polynomial with coefficients in A.

Modulo I, the equations say $x = y = 0$: the solution is unique "above $a = b = 0$". The theorem turns this into an algebraic certificate. The proof below constructs s explicitly from p, q, except for finitely many choices that are only merely given.

2. The computational setting

Homogenise: $P(X, Y, Z) = Z^d p(X/Z, Y/Z)$ and $Q = Z^e q(X/Z, Y/Z)$, so

$$P = XZ^{d-1} + aF + bG, \qquad Q = YZ^{e-1} + aU + bV.$$

Laurent polynomials with sign conditions. Let $\Lambda(n)$ be the Laurent polynomials in X, Y, Z over A, homogeneous of degree n (exponents in $\mathbb{Z}$). For a set E of variables, $\Lambda_E(n)$ is the part where only the variables in E may have negative exponents. For example $\Lambda_Z(0)$ consists of the $\sum c_{ij} X^i Y^j Z^{-i-j}$ with $i, j \geq 0$, i.e. of the polynomials in X/Z and Y/Z: it is $A[x, y]$ in disguise.

Triples. A 0-*triple* of degree n is $h = (h_X, h_Y, h_Z)$ with $h_X \in \Lambda_X(n)$, $h_Y \in \Lambda_Y(n)$, $h_Z \in \Lambda_Z(n)$. A 1-*triple* is (w_{XY}, w_{XZ}, w_{YZ}) with $w_E \in \Lambda_E(n)$. Put

$$\delta_0 h = (h_Y - h_X,\ h_Z - h_X,\ h_Z - h_Y), \qquad \delta_1 w = w_{XY} - w_{XZ} + w_{YZ} \in \Lambda_{XYZ}(n),$$

so that $\delta_1 \delta_0 = 0$. Two triples are *congruent*, $h \equiv h'$, when $h - h' = Pu + Qv$ componentwise, with u, v triples of the right shape and degrees $n - d$, $n - e$.

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Sections. A 0-triple h is a *section* if $\delta_0 h \equiv 0$: its three components agree, modulo the equations, on each overlap. Sections of degree 0 form a ring under componentwise product, since $(hh')_Z - (hh')_X = h_Z(h'_Z - h'_X) + h'_X(h_Z - h_X)$. The constant $(1,1,1)$ is a section (`CechSections`).

Restriction. For a 0-triple h of degree 0, $\mathrm{res}(h) \in L$ is obtained from h_Z by setting $X/Z = x$, $Y/Z = y$. It is additive and multiplicative, and kills $Pu+Qv$ because $p(x,y) = q(x,y) = 0$. For h of degree -1, $\mathrm{res}(Zh)$, $\mathrm{res}(Xh)$, $\mathrm{res}(Yh)$ are defined, and $\mathrm{res}(Xh) = x\,\mathrm{res}(Zh)$, $\mathrm{res}(Yh) = y\,\mathrm{res}(Zh)$, because $X = (X/Z)Z$.

So the plan is: *find a section h of degree -1 with* $\mathrm{res}(Zh) = 1 + ar + bt$, then show that restrictions of sections are integral over A.

3. The proof in five steps

Step 1: an almost-section. Let $z^* = (0, 0, Z^{-1})$, of degree -1. Then $\mathrm{res}(Zz^*) = 1$, but $\delta_0 z^* = (0, Z^{-1}, Z^{-1})$, which is not $\equiv 0$: z^* is not a section. It is, however, a section "up to arbitrarily high powers of I". On the overlap XZ, where X may be inverted, put $\theta = X^{-1}Z^{1-d}$ and $Ef = -\theta(aF + bG)f$. Since $P = XZ^{d-1} + aF + bG$, we get $f = Ef + \theta P f$, hence

$$Z^{-1} = E^n Z^{-1} + P\,u_n, \qquad E^n Z^{-1} \in I^n \Lambda_{XZ}(-1).$$

This is the Laurent form of a simple fact: when x is invertible, $p = 0$ gives $1 = a(-f/x) + b(-g/x) \in I$. The same holds on YZ with Q. Hence $\delta_0 z^* \equiv w_n$ with $w_n \in I^n$ for every n (`FixedCochain`).

Step 2: what stands between w_n and δ_0 of something small. If w_n were $\delta_0 c_n$ with $c_n \in I^n$, then $z^* - c_n$ would be a section with the same restriction as z^* modulo I^n, and we would be done. This is true only up to finitely many *obstructions*. Every Laurent polynomial q of Λ_{XYZ} decomposes as $q = \pi(q) + \delta_1\sigma(q)$. Here $\pi(q)$ keeps the monomials whose three exponents are all negative. Each other monomial lies in some Λ_E with two variables, and σ puts it there. If $\delta_1 w = 0$, then $w = \delta_0\tau(w)$ for an explicit τ (`CechH1Finite`, `CechContraction`). Applying this to w_n gives

$$\delta_0 z^* \equiv G\rho_n + \delta_0 c_n, \qquad c_n \in I^n, \quad \rho_n \in A^r \text{ with entries in } I^{n-1},$$

where $G\rho = \sum_t \rho_t m_t$ for fixed 1-triples $m_1, \dots, m_r$, one for each monomial with three negative exponents in a fixed degree (`CechF1`). So the only obstruction is the vector $\rho_n \in A^r$.

Step 3: the obstructions that can be removed form a finitely generated module. Let

$$N = \{\rho \in A^r \mid G\rho \equiv \delta_0 v \text{ for some } v\}.$$

By Step 2, $\rho_n \in N$, with $v = z^* - c_n$. Deciding whether $G\rho \equiv \delta_0 v$ has a solution amounts to linear equations over A on finitely many coefficients. The only nonlinear input is the description of the pairs (α, β) with $\alpha P + \beta Q = 0$: writing $P = D_0 P'$, $Q = D_0 Q'$ with $D_0 = \gcd(P, Q)$, they are $\alpha = -\gamma Q'$, $\beta = \gamma P'$ (the gcd hypothesis). This expresses N as the projection of the solutions of a finite matrix $\mathcal{M}$ over A. By (C), N is merely generated by $u_1, \dots, u_p$, with explicit b_j such that $Gu_j \equiv \delta_0 b_j$ (`ACLRelations`).

Step 4: the correction, by Artin–Rees. Take c from Theorem 1 for $N = \mathrm{span}(u_j) \subseteq A^r$, and $n = c + 2$. Then ρ_n has entries in I^{c+1} and lies in N, so $\rho_n = \sum_j \lambda_j u_j$ with $\lambda_j \in I$. Put

$$h = z^* - c_n - \sum_j \lambda_j b_j.$$

Then $\delta_0 h \equiv G\rho_n - \sum_j \lambda_j G u_j = 0$, so h is a section. Moreover $h - z^* \in I$, i.e. $h = z^* + am_1 + bm_2$, and so

$$s = \mathrm{res}(Zh) = 1 + a\,\mathrm{res}(Zm_1) + b\,\mathrm{res}(Zm_2) = 1 + ar + bt, \qquad \mathrm{res}(Xh) = sx, \quad \mathrm{res}(Yh) = sy$$

(`ACLCorrection`, `ACLCertificate`).

This is the step that needs Artin–Rees. Step 3 alone writes ρ_n as $\sum \mu_j u_j$ with arbitrary μ_j. Then h is still a section, but $h - z^*$ is no longer in I, and s need not be $\equiv 1$ modulo I.

Step 5: restrictions of sections are integral. Every section c of degree 0 is congruent to $\Phi(\kappa(c)) = \kappa_0(c) \cdot 1 + \sum_i \kappa_i(c) C_i$, for fixed sections C_i. Here $\kappa(c) \in A^{s'}$ is computed from c by the same π, σ, τ (`CechSections`). Apply the following lemma to $f_n = \kappa(c^n)$ in $A^{s'}$.

Lemma 3 (dependence)**.** *If A is CFN, every sequence $f_0, f_1, \ldots$ in $A^{s'}$ merely has a term $f_N \in \operatorname{span}(f_0, \ldots, f_{N-1})$ (`SeqDependence`).*

This is noetherianity of $A^{s'}$ in its weakest form: the spans cannot grow forever. Membership is decided by (D). From $\kappa(c^N) = \sum_{i<N} x_i \kappa(c^i)$ and linearity of Φ we get $c^N \equiv \sum_i x_i c^i$. Restricting, $\operatorname{res}(c)$ is a root of the monic polynomial $T^N - \sum_i x_i T^i$. Applied to $c = Zh, Xh, Yh$, this proves Theorem 2.

4. Remarks

- Noetherianity of A is used exactly twice: through Artin–Rees in Step 4, and through the dependence lemma in Step 5.
- Every "merely" (the generators u_j, the bound c, the coefficients λ_j, the index N) is eliminated into the conclusion, which is itself a proposition. No choice principle and no excluded middle are used.
- For readers who know the vocabulary: sections of degree n modulo $\equiv$ are the global sections of $\mathcal{O}_W(n)$ on $W = \{P = Q = 0\}$ in the projective plane over A. The obstructions of Step 2 are the classes in H^1, and Step 5 is the finiteness of H^0. None of this is used in the proof.